



Monte-Carlo ray tracing simulation of view factors for diffuse solar radiation determination in cylindrical photobioreactor arrays for microalgae cultivation

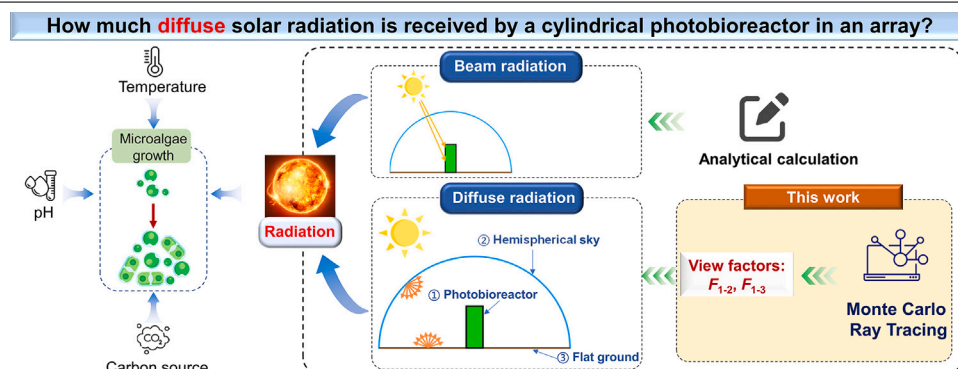
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GRAPHICAL ABSTRACT



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ABSTRACT

In this study, a Monte Carlo ray tracing (MCRT) model was developed to calculate view factors between cylindrical photobioreactor (PBR) sidewalls and sky/ground surfaces in outdoor array configurations. The effects of PBR dimension, inter-PBR spacing and array layout, namely square, staggered and biomimetic layouts, were analyzed. Results showed that view factors are mainly governed by mutual shading effects. In staggered arrays, centrally positioned PBRs exhibited significantly lower view factors compared to peripheral units, with reductions reaching up to 97%. Reducing inter-PBR spacing from 1.26 m to 0.56 m in staggered configurations resulted in a 46% decrease in mean view factor (from 0.301 to 0.163). Comparative analysis revealed average view factors of 0.2410, 0.2416 and 0.2259 for square, staggered, and biomimetic layouts, respectively. This work establishes a methodology for quantifying diffuse radiation received by PBR arrays which is indispensable for PBR multiphysics simulations.

1. Introduction

Microalgae cultivation has gained prominence for sustainable resource production and environmental benefits (Zhuang et al., 2018).

These photosynthetic microorganisms efficiently convert CO₂ and sunlight into biomass through various applications (Yoon et al., 2025; Lim et al., 2024; Wan Mahari et al., 2022), serving as renewable

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Nomenclature**Abbreviations**

2D	two-dimensional
MCRT	Monte-Carlo ray tracing
PAR	photosynthetically active radiation
PBR	photobioreactor
SOFC	solid-oxide fuel-cell

Greek Letters

λ	wavelength of light
σ	Stefan–Boltzmann constant
θ	angular distance between adjacent cylinders
θ_i	polar angle of the i th PBR in the biomimetic layout
ε	emissivity
φ	golden ratio

Latin Letters

$\bar{F}$	average view factor
$\dot{Q}$	diffuse radiative flux (W m^{-2})
$\dot{Q}_e$	radiative power
A	surface area (m^2)
a	center-to-center distance between adjacent PBRs in a staggered array in x direction (m)
b	center-to-center distance between adjacent PBRs in a staggered array in y direction (m)
E	emissive power
F	view factor
f	frequency function
g	probability density function
h	height of PBRs (m)
i	sequence of PBR in biomimetic layout
I_b	blackbody intensity
m	coefficient for adjusting column spacing
n	coefficient set as 0.5
N_{cv}	simulated cylindrical PBRs counter
N_{c}	number of simulation cylindrical PBRs
N_{rays}	number of simulation rays
N_{rv}	simulated rays counter
Q	diffuse radiation (W)
r	radius of PBRs (m)
r_i	polar radius of the i th PBR in the biomimetic layout (m)
s	clearance between adjacent cylindrical PBR side surfaces (m)

Subscripts

1	cylinder side surface
2	hemispherical sky
3	flat ground
c-e	central PBR side surface to each surrounding PBR side surface
v	model validation

with precise control of pH, temperature, and CO_2 concentrations, compared to open systems (Clagnan et al., 2023; Li et al., 2022). Cylindrical PBRs are widely used for microalgae cultivation because of their compact structure, modularity, easy operation, and excellent gas–liquid transfer performance (Xu et al., 2009). Advanced PBR designs have accelerated productivity improvements and commercialization efforts (Huang et al., 2017; Wolf et al., 2016; Barboza-Rodríguez et al., 2024).

The majority of the state-of-the-art experimental and modeling studies of cylindrical microalgae PBRs discussed the underlying physics of a single PBR (Béchet et al., 2013, 2014; Loubière et al., 2009). In arrays, shading effects and diffuse radiation from the sky/ground significantly alter irradiance distribution, complicating radiation modeling, which is critical for system optimization. Among existing studies, Huang et al. (2019) employed the canyon effect to estimate the diffuse light received by reactors when designing a predictive biomass productivity model for large-scale cylindrical PBR arrays. However, this method for calculating incident diffuse beam is computationally intensive and time-consuming. Alternatively, the diffuse radiation absorbed by a PBR can be indirectly calculated using the view factor, which quantifies the proportion of radiation emitted from one diffuse reflecting surface that is transmitted to another (Holman, 2010; Modest and Mazumder, 2013). The view factor, which governs the proportion of diffuse light intercepted by PBR surfaces, plays a critical role in regulating light intensity distribution and consequently influences microalgal growth dynamics. Excessive localized irradiance due to improper view factor calculations may induce photoinhibition and reduce photosynthetic efficiency, while optimized diffuse light allocation promotes cultivation performance. Unfortunately, there are no analytical solution for the view factor from the hemispherical sky or the flat ground to the side wall of cylindrical reactors, which is a key factor limiting the development and function of the PBR numerical simulation. For instance, due to this research gap, in the multiphysics model previously developed for concentric-tube internal loop airlift PBRs for microalgae cultivation, Li et al. (2023) assumed the view factor from the sky to the PBR as 0.5 and neglected the diffuse reflected light from the ground, which may lead to inaccuracy in the simulated results and conclusions.

Monte Carlo ray tracing (MCRT) emerges as a promising solution, simulating beam propagation via random sampling to accurately compute view factors in complex arrays. The core concept of the MCRT method involves simulating the propagation of beams within a real-world scene. In this simulation, the thermal radiation that emanates from a surface is treated as a set of diffuse energy beams. Then, the radiation received by a target surface is determined based on the number of beams absorbed. This approach provides a robust framework for analyzing radiative transport in complex cylinder array configurations, making it a valuable tool for applications involving large-scale photobioreactor arrays (Creswell, 2010) or other energy systems with an array of cylindrical reactors (George et al., 2015; Kee et al., 2010; Wang et al., 2021). In industrial-scale outdoor PBR systems, designers face practical trade-offs between maximizing reactor density and minimizing mutual shading. Current empirical designs often rely on trial-and-error approaches due to the lack of quantitative guidelines for parameter optimization. This parametric study aims to bridge this gap by providing actionable thresholds to inform data-driven reactor arrangement decisions.

In particular, inspired by the advancements in sunflower inspired designs for optimizing performance and resource utilization, this study extends the concept to the layout optimization of cylindrical PBR arrays and proposes an innovative biomimetic sunflower arrangement. This novel design aims to optimize solar beam distribution within the array by mimicking nature patterns that enhance light capture efficiency. Alkhalidi et al. (2024) proposed a sunflower inspired urban planning pattern and demonstrated that, for rooftop areas with available sunlight, the sunflower layout outperformed the grid and radial patterns by 4%. Belaid et al. (2022) investigated the effect of heliostat field distribution on solar power tower systems and found that, for the PS

biofuel sources that reduce fossil fuel dependence and greenhouse emissions (Ganesh Saratale et al., 2022; Wang et al., 2022). Cultivation occurs in open ponds or closed photobioreactors (PBRs). PBRs are closed, controlled culture vessels that allow higher cell concentrations

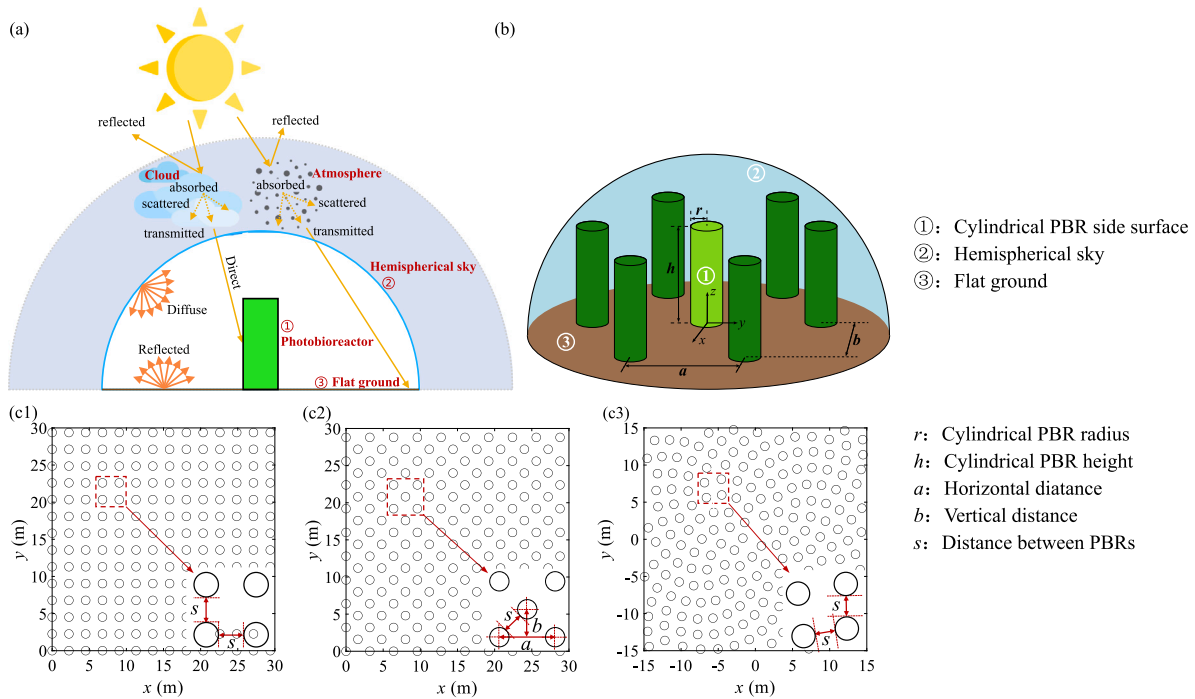


Fig. 1. Schematic diagrams illustrating (a) solar radiation incident on a cylindrical photobioreactor; (b) an example cylindrical photobioreactor array; and (c) three layout patterns of photobioreactors investigated in this study: (c1) square layout, (c2) staggered layout, and (c3) biomimetic layout. Key geometrical parameters include: horizontal and vertical center-to-center distances between adjacent reactors, a and b , respectively, clearance between reactor side surfaces, s , photobioreactor radius r , and height h . The studied surfaces are denoted as ①: cylindrical surfaces; ②: hemispherical sky; ③: flat ground.

10 plant installed in Spain, the biomimetic sunflower spiral field could reduce land area by 25.86% (approx. 113.538 m²).

In this study, the Monte Carlo ray tracing method is employed to numerically calculate the view factors between the side wall of cylindrical PBRs (arranged in an array configuration) and the hemispherical sky or the flat ground. This work demonstrates the effects of key geometrical parameters, including the PBR radius, height, and the spacing between adjacent PBRs, on the view factors. Moreover, an innovative biomimetic sunflower inspired layout for cylindrical PBRs is investigated and compared with traditional square and staggered layouts to explore the influences of reactor arrangement on the view factors. The findings of this study provide valuable insights into the optimal arrangement of PBRs for large-scale commercial microalgae cultivation. By improving light conditions suited to microalgae growth, the results serve as a practical guide for designing efficient outdoor PBR system, thereby contributing to enhancing productivity and scalability in bioengineer applications based on microalgae.

2. Model system

A schematic diagram of an outdoor large-scale PBR array for microalgae cultivation is shown in Fig. 1(b). The schematic illustrates a representative system comprising seven PBRs of the same radius r and the same height h . The cylindrical surface of the PBR at the center, the hemispherical sky and the flat ground are marked as surface 1, 2 and 3, respectively. The global coordinate system is constructed at the center of the base of the central PBR, with the positive x - and y -axes pointing along the South and the East directions, respectively. In this study, the PBR array is assumed to be arranged on a square land of the length and the width both equal to 30 m.

The PBRs are arranged in the field in defined patterns. Three types of PBR layout patterns are discussed in this work: (1) square layout (Fig. 1c1), (2) staggered layout (Fig. 1c2) and (3) biomimetic layout (Fig. 1c3). For the square layout, the PBRs are arranged in parallel along both x and y axes and adjacent PBRs have an equal surface-to-surface

distance of s . In a staggered layout, the spacings along x and y axes between adjacent reactor centers are denoted as a and b , respectively. The surface-to-surface distance between neighboring cylindrical reactors is s . The PBRs are arranged with an equal center-to-center distance between adjacent reactors along x and y directions. This configuration ensures efficient space utilization and better light distribution across the array. Inspired by the employment of the sunflower layout in energy systems, the outdoor PBRs arranged in the biomimetic sunflower layout pattern is explored in this work. The positions of the elements are described as the following (Vogel, 1979):

$$\theta_i = 2\pi\varphi^{-2}i \quad (1)$$

$$r_i = mi^n \quad (2)$$

where θ_i is linearly related to the element sequence i and r_i is a radial growth function proportional to the constant exponentiation of i , and the angular component φ is equal to the golden ratio $\frac{1+\sqrt{5}}{2}$. In this work, the coefficient n was defined to be 0.5, at which point the mean distance between neighboring cylindrical photobioreactors in the biomimetic layout is constant (Ridley, 1982), and the spacing distance of the columns is adjusted by the coefficient m .

Solar attenuation — the reduction of solar radiation through atmospheric absorption, scattering, and reflection by clouds and gases — critically governs the radiative energy available to cylindrical PBRs in large-scale outdoor arrays. The incident radiation on PBR surfaces is composed of three components (see Fig. 1a): (1) beam solar radiation, (2) diffuse sky radiation that is scattered by the atmospheric molecules, aerosols and clouds, and (3) diffuse ground-reflected radiation. Although the beam radiation dominates under clear skies, diffuse radiation contributes significantly to the total irradiance, particularly under cloudy conditions or in densely packed reactor arrays where mutual shading reduces direct exposure. The geometrical view factor, which quantifies the fraction of radiation exchanged between surfaces, plays a pivotal role in determining the spatial distribution of diffuse radiation on PBR surfaces. In PBR arrays, the view factor between

reactors and their surroundings (e.g., sky, ground, and adjacent reactors) governs the magnitude of both atmospheric and ground-reflected diffuse radiation. This is especially critical for microalgae cultivation, as the organism's growth kinetics depend nonlinearly on light intensity, and heterogeneous light distribution can lead to suboptimal productivity or photoinhibition. By explicitly modeling the view factor in radiative transfer analyses, the interplay between reactor geometry, array configuration, and diffuse radiation capture can be optimized. Such insights are essential for designing energy-efficient PBR systems that maximize light utilization while mitigating losses from shading and attenuation.

3. Methodology

3.1. View factor

According to the definition of view factor (Modest and Mazumder, 2013), if the diffuse sky and the ground-reflected radiative fluxes are denoted as $\dot{Q}_2$ and $\dot{Q}_3$ (W m^{-2}), respectively, the diffuse radiation received by the PBR side surface from the sky and the ground, Q_{2-1} (W) and Q_{3-1} (W), can be calculated as: $Q_{2-1} = \dot{Q}_2 A_2 F_{2-1}$, $Q_{3-1} = \dot{Q}_3 A_3 F_{3-1}$. F_{2-1} and F_{3-1} are the view factors from the sky and the ground to the cylindrical side surface, i.e., the proportions of the energy emitted from the sky or the ground that is captured by the cylindrical side surface, A_2 (m^2) is the surface area of the hemispherical sky, A_3 (m^2) is the surface area of the flat ground. According to the reciprocity relation of the view factors, $A_2 F_{2-1} = A_1 F_{1-2}$, $A_3 F_{3-1} = A_1 F_{1-3}$, then, $Q_{2-1} = \dot{Q}_2 A_1 F_{1-2}$, $Q_{3-1} = \dot{Q}_3 A_1 F_{1-3}$. F_{1-2} and F_{1-3} represent the view factors between the cylindrical side surface and the hemispherical sky and the view factor between the cylindrical side surface and the flat ground, respectively, and A_1 (m^2) is the area of the cylindrical side surface.

View factors of F_{1-2} and F_{1-3} are easier to calculate than F_{2-1} and F_{3-1} using MCRT technique, because the ray generation surface is transformed from the hemispherical sky or the ground to the cylindrical surface. Given that the sky and ground domains span orders of magnitude larger areas than the cylindrical PBR side surfaces, the required number of rays for numerical convergence is substantially reduced, leading to a significant reduction in computational costs.

3.2. Monte Carlo

The core principle of the Monte Carlo method involves employing random number generation and statistical simulation for numerical computation. It is assumed that radiative variables are independent of each other and position on the emitting surface element. The radiative power $d\dot{Q}_e(T, \lambda, \hat{s})$ emitted from a surface element dA , defined as the total radiative energy emitted from dA in all directions per unit time, can be expressed as:

$$\begin{aligned} d\dot{Q}_e(T, \lambda, \hat{s}) &= E'_\lambda(T, \lambda, \hat{s}) d\Omega d\lambda dA \\ &= \epsilon'_\lambda(T, \lambda, \theta, \varphi) E'_{b\lambda}(T, \lambda, \theta, \varphi) \sin \theta d\theta d\varphi d\lambda dA \\ &= \epsilon'_\lambda(T, \lambda, \theta, \varphi) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta d\theta d\varphi d\lambda dA \end{aligned} \quad (3)$$

the frequency function for surface emission can be expressed as:

$$\begin{aligned} f_e(T, \lambda, \theta, \varphi) &= \frac{d\dot{Q}_e(T, \lambda, \theta, \varphi)}{d\lambda d\theta d\varphi} \\ &= \epsilon'_\lambda(T, \lambda, \theta, \varphi) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta dA \end{aligned} \quad (4)$$

the probability density function for surface emission can be calculated as:

$$\begin{aligned} g_e(\lambda, \theta, \varphi) &= \frac{\epsilon'_\lambda(T, \lambda, \theta, \varphi) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta}{\int_{\lambda=0}^{\infty} \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \epsilon'_\lambda(T, \lambda, \theta, \varphi) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta d\theta d\varphi d\lambda} \\ &= \frac{\epsilon'_\lambda(T, \lambda, \theta, \varphi) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta}{\epsilon(T) \sigma T^4} \end{aligned} \quad (5)$$

the probability of emission into $d\lambda d\theta d\varphi$:

$$dP_e(\lambda \leq \lambda \leq \lambda + d\lambda, \theta \leq \theta \leq \theta + d\theta, \varphi \leq \varphi \leq \varphi + d\varphi) = g_e(\lambda, \theta, \varphi) d\lambda d\theta d\varphi \quad (6)$$

Furthermore, random values are generated within the interval [0, 1] instead of the cumulative distribution functions, as expressed below:

$$R_\lambda = F_{e,\lambda} = \int_{\lambda^*=0}^{\lambda} \left[\int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi/2} g_e(\lambda^*, \theta, \varphi) d\theta d\varphi \right] d\lambda^* \quad (7)$$

$$R_\theta = F_{e,\theta} = \int_{\theta^*=0}^{\theta} \left[\int_{\varphi=0}^{2\pi} \int_{\lambda=0}^{\infty} g_e(\lambda, \theta^*, \varphi) d\lambda d\varphi \right] d\theta^* \quad (8)$$

$$R_\varphi = F_{e,\varphi} = \int_{\varphi^*=0}^{\varphi} \left[\int_{\theta=0}^{\pi/2} \int_{\lambda=0}^{\infty} g_e(\lambda, \theta, \varphi^*) d\lambda d\theta \right] d\varphi^* \quad (9)$$

In this study, the light rays are emitted from the cylindrical PBR side surface. The specific emission position is determined by generating two random numbers R_ϕ and R_z , which correspond to the angular position ϕ and the vertical position z , respectively.

$$R_\phi = F_\phi(\phi) = \frac{\int_{\phi^*=0}^{\phi} \int_{z=0}^h \epsilon(T) \sigma T^4 dz d\phi^*}{\int_{\phi=0}^{2\pi} \int_{z=0}^h \epsilon(T) \sigma T^4 dz d\phi} = \frac{\phi}{2\pi} \quad (10)$$

$$R_z = F_z(z) = \frac{\int_{z^*=0}^z \int_{\phi=0}^{2\pi} \epsilon(T) \sigma T^4 d\phi dz^*}{\int_{z=0}^h \int_{\phi=0}^{2\pi} \epsilon(T) \sigma T^4 d\phi dz} = \frac{z}{h} \quad (11)$$

Hence,

$$\begin{cases} x = r \cos \phi = r \cos(2\pi R_\phi) \\ y = r \sin \phi = r \sin(2\pi R_\phi) \\ z = h R_z \end{cases} \quad (12)$$

The direction of emitted rays is determined using two random numbers, R_θ and R_φ (varied in the range [0, 1]). These random numbers are used to define the angular coordinates specifying the emission direction in a spherical coordinate system.

$$R_\theta = \frac{\int_{\theta^*=0}^{\theta} \left(\int_{\varphi=0}^{2\pi} \int_{\lambda=0}^{\infty} \epsilon'_\lambda(T, \lambda, \theta^*, \varphi) I_{b\lambda}(T, \lambda) \cos \theta^* \sin \theta^* d\lambda d\varphi \right) d\theta^*}{\epsilon(T) \sigma T^4} \quad (13)$$

$$R_\varphi = \frac{\int_{\varphi^*=0}^{\varphi} \left(\int_{\theta=0}^{\pi/2} \int_{\lambda=0}^{\infty} \epsilon'_\lambda(T, \lambda, \theta, \varphi^*) I_{b\lambda}(T, \lambda) \cos \theta \sin \theta d\lambda d\theta \right) d\varphi^*}{\epsilon(T) \sigma T^4} \quad (14)$$

The PBR side surface are assumed to be the diffuse-gray surface, and θ and φ are independent of each other. Hence,

$$\begin{cases} R_\theta = \sin^2 \theta \\ R_\varphi = \frac{\varphi}{2\pi} \end{cases} \quad (15)$$

$$\begin{cases} u_x = \sin \theta \cos \varphi = \sin \left(\sin^{-1} \sqrt{R_\theta} \right) \cos(2\pi R_\varphi) \\ u_y = \sin \theta \sin \varphi = \sin \left(\sin^{-1} \sqrt{R_\theta} \right) \sin(2\pi R_\varphi) \\ u_z = \cos \theta = \cos \left(\sin^{-1} \sqrt{R_\theta} \right) \end{cases} \quad (16)$$

3.3. Computational techniques

In this work, a Monte Carlo ray tracing program written in FORTRAN 90 is in-house developed to compute the view factors. The program is designed to simulate the propagation of beam within the cylindrical PBR array and compute the corresponding view factors from the reactor side walls to the hemispherical sky or the flat ground. The detailed computation procedure is described in Fig. 2. The workflow outlines the MCRT method employed to calculate the view factors (F_{1-2} and F_{1-3}). The key parameters of the MCRT model include ray number and geometrical parameters (see supplementary material). Key steps include: (1) geometrical projection of radiation sources (PBR side surfaces) onto the sky and ground; (2) stochastic emission of

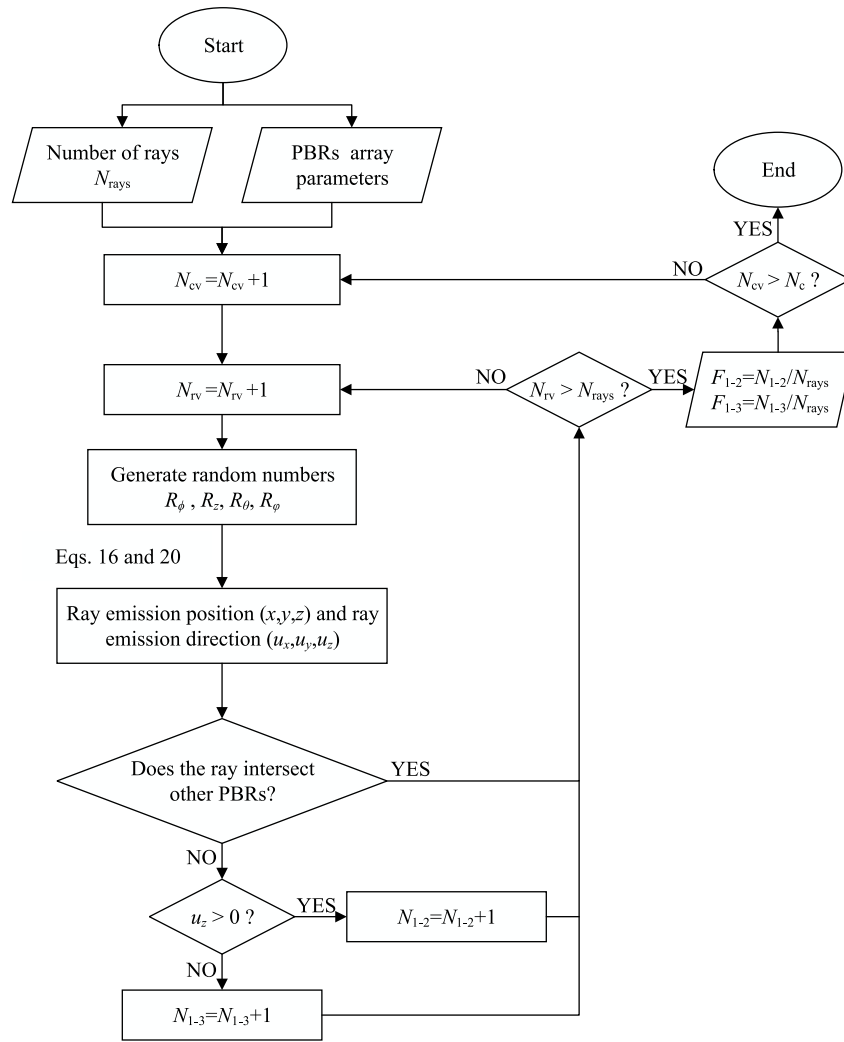


Fig. 2. Computational flowchart for determining the view factor between the cylindrical photobioreactor side surface and the hemispherical sky, F_{1-2} , and the view factor between the cylindrical photobioreactor side surface and the flat ground, F_{1-3} .

rays from the PBR surface with angular distributions weighted by the Lambert's cosine law; (3) ray tracing to determine the intersection points of rays with the sky or ground domains; (4) statistical tallying of intercepted rays to compute view factors via $F_{1-2} = N_{1-2}/N_{rays}$ and $F_{1-3} = N_{1-3}/N_{rays}$, where N_{1-2} and N_{1-3} are the number of rays reaching surfaces 2 (the sky) and 3 (the ground), respectively; and (5) iterative refinement until convergence criteria are met. The computational efficiency arises from the sky and ground domains' large spatial extent relative to the PBR, minimizing ray counts required for statistical accuracy.

4. Results and discussion

In this section, the results obtained from the Monte Carlo ray tracing simulations are presented and analyzed to understand the influence of key geometrical parameters on the view factors of cylindrical PBRs. The effects of reactor radius, height, and spacing on the radiative exchange between the PBR sidewalls and the surrounding environment, including the hemispherical sky and flat ground, are discussed in detail. Furthermore, the performance of different reactor arrangements, including square, staggered, and the biomimetic sunflower-inspired layout, is compared to evaluate their impact on view factors. These findings offer valuable insights for optimizing reactor configurations to enhance light availability and improve the efficiency of large-scale microalgae cultivation.

4.1. Ray number independence study

The Monte Carlo method is a statistical numerical technique based on probability theory, where the accuracy of the results improves as the sample size increases. In MCRT simulations, the number of rays generated directly influences the accuracy and convergence of the results. The results illustrate the variation in the view factor from the side surface of a cylindrical PBR located at the center of a staggered layout to the sky as the number of rays, N_{rays} , increases, with baseline parameters: $r = 0.5$ m, $h = 1.6$ m, $a = 6$ m, $b = 3$ m (see supplementary material). It is found that the view factor gradually converges as the number of rays increases, and when the number of rays reaches a certain threshold, the change in the view factor becomes negligible, stabilizing at a value close to the true value. In other words, once the number of rays surpasses a specific threshold, further increases have minimal impact on the view factor, indicating that the simulation has achieved sufficient accuracy. In this study, to ensure the accuracy and reliability of the results, 10^7 rays are generated per simulation. This approach ensures that the simulation results are sufficiently close to the actual values, thereby enhancing both the precision and credibility of the calculations.

4.2. Model validation

In order to verify the correctness of the developed program, two cases of geometrical configurations where the view factors can be

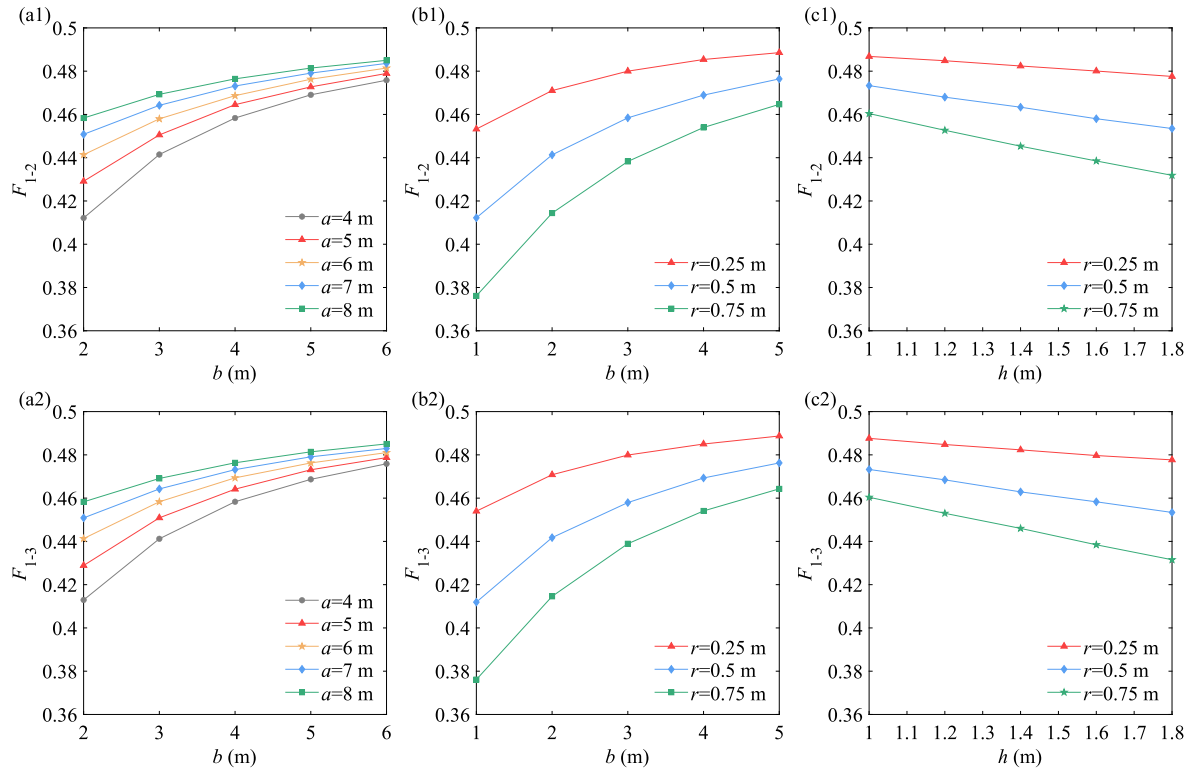


Fig. 3. The change of (1) the view factor between the central cylindrical side surface and the hemispherical sky, F_{1-2} , and (2) the view factor between the central cylindrical side surface and the ground, F_{1-3} , as functions of: (a) the horizontal and vertical spacings between adjacent photobioreactors, a and b , respectively; (b) the vertical spacing between adjacent photobioreactors, b , and the photobioreactor radii, r ; (c) the height of photobioreactors, h , and the photobioreactor radii, r . When one or two variables are assessed, the other variables are taken as the baseline values listed in Table 3.

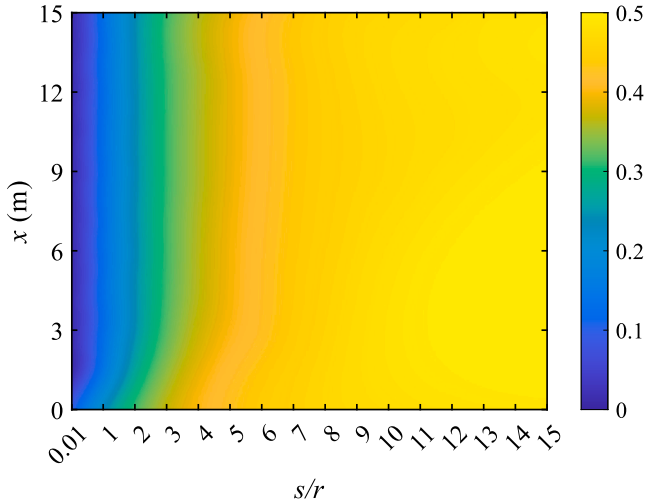


Fig. 4. Spatial distribution of view factor from photobioreactor side surface and the sky, F_{1-2} , in a staggered array. The contour map quantifies F_{1-2} for cylindrical photobioreactors within a fixed 30m×30m cultivation area. The dimensionless gap-to-radius ratio (s/r) is modulated by varying the surface-to-surface spacing s between adjacent reactors while maintaining a constant radius r (of the baseline value 0.5 m). Color intensity corresponds to F_{1-2} values, with the horizontal axis (x) representing reactor positions along the central row.

calculated analytically are employed in this study: (1) infinitely long parallel cylinders of the same diameter (see supplementary material), and (2) finite equal-length nested cylinders (see supplementary material). By comparing the numerical solutions calculated by the MCRT program with the analytical solutions, the accuracy and rationality of the program can be evaluated effectively. Analytical expressions for

Table 1

Comparison of view factors and relative errors for infinitely long parallel cylinders of the same diameter at varying H/r ratios.

H/r	View factor	Relative error
2×10^1	0.0809798	−6.46%
2×10^2	0.0861925	−0.44%
2×10^3	0.0867546	0.21%
2×10^4	0.0867219	0.17%
2×10^5	0.0865353	−0.05%
2×10^6	0.0865862	0.01%
infinite length	0.0865746	–

Table 2

View factors and relative errors for finite equal-length nested cylinders at different angular distances θ .

θ	View factor	Relative error
1°	0.5655533	0.37%
2°	0.5650154	0.46%
3°	0.5642293	0.60%
4°	0.5647730	0.51%
5°	0.5619435	1.01%
6°	0.5604286	1.29%
Theoretical Value	0.5676370	–

the view factors of the classical models can be found in Modest and Mazumder (2013).

For the first configuration, the view factor between the side surfaces of two infinitely long parallel cylinders of a radius r and a surface-to-surface distance s , $F_{v,1-2}$, can be calculated as:

$$F_{v,1-2} = \frac{1}{\pi} (\sin^{-1} \frac{1}{X} + \sqrt{X^2 - 1} - X) \quad (17)$$

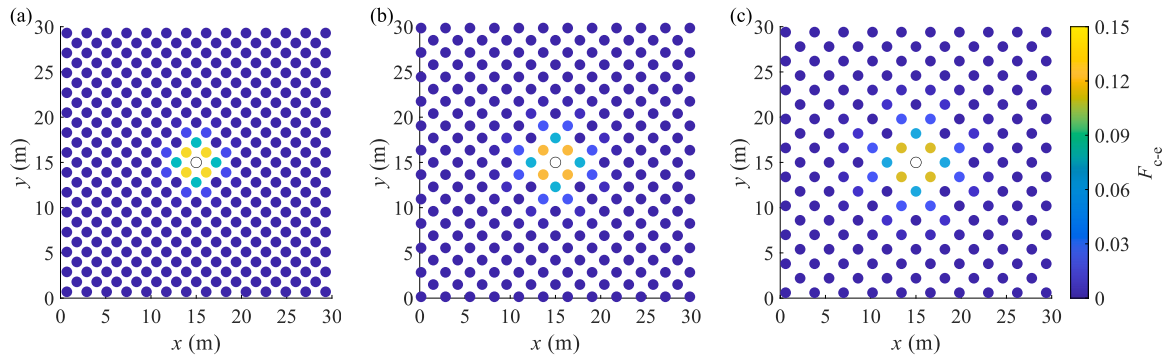


Fig. 5. View factors between the central photobioreactor and the surrounding photobioreactors, F_{c-s} , for systems with three surface-to-surface spacings: (a) $s = 0.56$ m, (b) $s = 0.91$ m, (c) $s = 1.26$ m, in a staggered layout with baseline photobioreactor geometrical parameters of $r = 0.5$ m and $h = 1.6$ m.

Table 3

Baseline and simulated parameters.

Baseline parameters	Simulated parameters
$r = 0.5$ m, $h = 1.6$ m	$a = 4$ m, 5 m, 6 m, 7 m, 8 m $b = 2$ m, 3 m, 4 m, 5 m, 6 m
$a = 6$ m, $h = 1.6$ m	$r = 0.25$ m, 0.5 m, 0.75 m $b = 1$ m, 2 m, 3 m, 4 m, 5 m
$a = 6$ m, $b = 3$ m	$r = 0.25$ m, 0.5 m, 0.75 m $h = 1$ m, 1.2 m, 1.4 m, 1.6 m, 1.8 m

$$X = 1 + \frac{s}{2r} \quad (18)$$

The cylinders are infinitely long, which is impossible to achieve in practical numerical calculation. Thus, $F_{v,1-2}$ for selected values of H/r varying from 20 to 2×10^6 are computed. As shown in Table 1, the relative error decreases rapidly as H/r increases, and it is reduced to 0.01% when H/r is equal to 2×10^6 . These results suggest a good agreement between the two methods.

For the second configuration, the view factor between the interior surface of the outer cylinder and the exterior surface of the inner cylinder, $F_{v,2-1}$, can be analytically calculated as:

$$F_{v,2-1} = \frac{1}{R} \left(1 - \frac{H^2 + R^2 - 1}{4H} - \frac{1}{\pi} \left\{ \cos^{-1} \frac{H^2 - R^2 + 1}{H^2 + R^2 - 1} - \frac{\sqrt{(H^2 + R^2 + 1)^2 - 4R^2}}{2H} \cos^{-1} \frac{H^2 - R^2 + 1}{R(H^2 + R^2 - 1)} - \frac{H^2 - R^2 + 1}{2H} \sin^{-1} \frac{1}{R} \right\} \right) \quad (19)$$

where,

$$R = \frac{r_2}{r_1} \quad (20a)$$

$$H = \frac{h}{r_1} \quad (20b)$$

Since ray transmission in the cylinder array is the core to be verified, in the MCRT simulation, the outer cylinder is replaced by an array of cylinders that are tangent to each other and their centers are evenly spaced with an angular distance of θ on the circumference of the outer cylinder. When the radius of the cylinders remains invariant, the gap between the surrounding cylinders exhibits a decreasing trend as θ diminishes. This reduction in the gap leads to a decrease in the number of rays that can pass through it. It is identified as the primary error source. As shown in Table 2, the relative error increases gradually with increasing θ . The relative error is equal to 0.368% when $\theta = 1^\circ$. Hence, the developed MCRT model provides sufficient precision for calculating the view factors required in this work.

4.3. Parametric analysis of photobioreactor geometry and spacing on radiative exchange

Using the developed MCRT algorithm, the influence of geometrical parameters on the radiative exchange between a cylindrical PBR array and the sky and the flat ground is systematically evaluated. The view factors between the PBR surfaces and their radiative surroundings (hemispherical sky and flat ground, F_{1-2} and F_{1-3} , respectively) are computed. The parameters under investigation include the PBR radius r , height h , and the horizontal and vertical center-to-center spacing between adjacent reactors, a and b , respectively. In this study, two parameters are taken as the baseline values (see Table 3), while the remaining two are independently varied across defined ranges to isolate their effects. The analysis assumes a large-scale PBR array deployed within a 30 m × 30 m square outdoor field, approximating commercial cultivation conditions.

Figs. 3(a1) and (a2) illustrate the variation in view factors from the side surface of the central cylindrical PBR to the sky and ground, F_{1-2} and F_{1-3} , respectively, as functions of inter-reactor spacing in a staggered layout with fixed reactor dimensions. The results reveal a monotonic increase in view factors with greater spacing, although the growth rate diminishes at larger distances. This behavior stems from reduced mutual shading between adjacent PBRs as spacing parameters a and b increase, enabling a larger fraction of radiant energy emitted from the central PBR's side surface to escape to the sky and ground. Beyond a critical spacing threshold, however, the marginal improvement in view factors becomes negligible, suggesting saturation in shading reduction. The interplay between reactor radius and spacing is examined in Figs. 3(b1) and (b2). For a fixed inter-reactor distance, increasing the PBR radius correlates with reduced view factors. This inverse relationship arises from enhanced occlusion by neighboring reactors: larger radii amplify geometrical shading, thereby limiting the effective radiative exchange area between the central PBR and its surroundings. This result highlights a critical design trade-off: while larger reactors reduce infrastructure costs (fewer units per area), they exacerbate shading losses. Figs. 3(c1) and (c2) explore the dependence of view factors on PBR radius r and height h under constant inter-reactor spacing. The view factors exhibit an inverse relationship with reactor height, as taller PBRs intensify mutual shading within the fixed-spacing array. This geometric constraint reduces the radiative flux escaping to both the sky and ground, highlighting the competing effects of reactor scale and spatial arrangement on light distribution.

Collectively, these results demonstrate that mutual shading governs the sensitivity of view factors to PBR dimensions and spacing. Minimal inter-reactor shading — achieved through smaller reactor sizes or larger spacings — maximizes radiative exchange with the sky and ground, yielding higher view factors. In particular, the analysis reveals symmetry in the system: the view factor from the central PBR sidewall to the hemispherical sky is equal to that of the flat ground (see Fig. 3). This

equivalence justifies the subsequent focus on sky-directed view factors in Sections 4.4–4.5 for parametric optimization. Across the parameter space investigated, the sky-directed view factor varies between 0.376 (minimized at $r = 0.75$ m, $h = 1.6$ m, $a = 6$ m, $b = 1$ m) and 0.489 (maximized at $r = 0.25$ m, $h = 1.6$ m, $a = 6$ m, $b = 5$ m). The difference of view factors between these extrema underscores the significant yet bounded influence of geometrical parameters on radiative efficiency in staggered PBR arrays.

In addition, Fig. 4 quantifies the spatial variation in the view factor to the sky, F_{1-2} , for PBRs along the central row of a staggered array, which systematically evaluates the impact of the dimensionless gap-to-radius ratio (s/r). The results reveal two critical trends: (1) Positional dependence: The centrally located PBR ($x = 15$ m) with the minimal inter-reactor spacing simulated ($s/r = 0.01$) exhibits the lowest view factor ($F_{1-2} = 0.0146$), attributable to the intense mutual shading from adjacent reactors. In contrast, the edge-positioned PBR ($x = 0$) with the maximal spacing ($s/r = 15$) achieves the highest F_{1-2} of 0.49, due to the reduced shading and the enhanced sky exposure. This spatial gradient implies that the edge reactors efficiently act as “light harvesters” due to the reduced occlusion, while the central reactors suffer from cumulative shading effects from the neighboring units—a critical consideration for balancing productivity across large-scale arrays. (2) Parametric sensitivity: For fixed s/r , increasing the horizontal coordinate x (i.e., moving toward the array center) reduces F_{1-2} by up to 97%, as reactors become progressively enveloped by neighboring units. For fixed x , increasing s/r initially elevates F_{1-2} due to diminished shading; however, beyond a critical threshold ($s/r \geq 11$), further increases yield negligible gains (<1%), asymptotically approaching a plateau ($F_{1-2} \approx 0.48$). These findings underscore a trade-off between spatial density and radiative efficiency in staggered PBR arrays. The saturation of F_{1-2} at $s/r \geq 11$ suggests an optimal spacing regime where additional separation no longer meaningfully improves light capture. This result provides a clear engineering guideline: expanding spacing beyond this threshold yields diminishing returns in light capture, making it economically unjustifiable.

4.4. Inter-reactor radiative coupling: View factors in photobioreactor arrays

Fig. 5 reveals an inverse relationship between inter-reactor spacing s and the view factor from the central PBR to its neighbors, F_{c-e} . At the minimal clearance studied (e.g., $s = 0.56$ m), the pronounced mutual shading elevates view factors between the central PBR and adjacent reactors due to geometrical occlusion. As spacing increases, this shading diminishes progressively, leading to a monotonic reduction in radiative coupling. The view factor exhibits radial decay, peaking for nearest neighbors (e.g., 0.1367 at $s = 0.56$ m) and approaching negligible values (0) for peripheral reactors in the array. Notably, array density governs both the magnitude and spatial distribution of radiative interactions. For denser configurations ($s = 0.56$ m), view factors are strongly localized, with 86% of the central PBR’s radiative exchange concentrated within its first two neighboring rings. Conversely, sparser arrays ($s = 1.26$ m) exhibit weaker but more uniform radiation distribution, as evidenced by the reduced nearest-neighbor view factor (0.1161) and broader angular dispersion. These findings emphasize that inter-reactor spacing not only dictates the magnitude of radiative coupling but also shapes its spatial footprint. For instance, in densely packed systems ($s = 0.56$ m), optimizing reactor placement to mitigate localized shading hotspots — particularly for central reactors — could enhance overall light utilization. Conversely, sparse configurations ($s = 1.26$ m) may prioritize uniformity, enabling more predictable light distribution across the array.

Table 4

Mean view factors between the photobioreactor side surface and the sky, $\bar{F}_{1-2}$, for photobioreactors arranged in different layout patterns with different values of photobioreactor radius r and spacing s .

PBR radius r	Layout pattern	$s = 0.56$ m	$s = 0.91$ m	$s = 1.26$ m
$r = 0.25$ m	Square layout	0.1584854	0.2549269	0.3200372
	Staggered layout	0.1558466	0.2539632	0.3217881
	Biomimetic layout	0.1526359	0.2483174	0.3085665
$r = 0.5$ m	Square layout	0.1634155	0.2410123	0.3008074
	Staggered layout	0.1633434	0.2415572	0.30008071
	Biomimetic layout	0.1344991	0.2258670	0.2842616
$r = 0.75$ m	Square layout	0.1975952	0.2621156	0.2996283
	Staggered layout	0.1960674	0.2594572	0.3020864
	Biomimetic layout	0.1777726	0.2424487	0.2917542

4.5. Comparative analysis of photobioreactor array configurations

While current research predominantly addresses the design of individual PBRs, systematic optimization of large-scale outdoor PBR arrays remains under-explored. This study evaluates the radiative efficiency of cylindrical PBR arrays by analyzing the average view factor between the PBR side surface and the sky, a critical determinant of photosynthetically active radiation (PAR) capture in microalgae cultivation. Three array configurations — square, staggered, and a biomimetic sunflower-inspired layout — are compared within a fixed 30 m × 30 m cultivation area. Parametric simulations assess the impact of inter-reactor spacing (s) and PBR radius (r) under diverse geometrical scenarios, ensuring robustness through MCRT convergence criteria and statistical averaging.

Table 4 presents the simulated view factors for selected cases of PBR radii and spacing and layout patterns. The total numbers of PBRs for the three layout patterns are kept close to exclude the effect of PBR average spacing.

Figs. 6(a1–a3) demonstrate that an increased s enhances F_{1-2} by reducing the mutual shading. For instance, doubling s from 0.5 m to 1.0 m elevates F_{1-2} by 27% (0.28 to 0.36), as wider spacing mitigates radiative obstruction between adjacent reactors.

Contrary to intuitive expectations, Figs. 6(b1–b3) reveal a non-monotonic relationship between r and F_{1-2} . While larger r intensifies shading per reactor, the reduced reactor density (e.g., 481 to 162 units as r increases from 0.25 m to 0.75 m) disproportionately elevates the system-wide average F_{1-2} by 15% (0.29 to 0.34). Maximizing F_{1-2} requires balancing s and r to reconcile land-use efficiency with radiative exposure. A spacing threshold ($s/r \geq 4$) ensures diminishing returns in shading reduction.

Figs. 6(c1–c3) demonstrates that all three PBR arrangements (square, staggered, and biomimetic) exhibit similar trends in average view factors from reactor sidewalls to the sky. As quantified in Table 4, the peak system-wide radiative efficiency occurs at $r = 0.25$ m and $s = 1.26$ m, with average view factors of 0.32, 0.322 and 0.309 for the square, staggered and biomimetic layouts, respectively. While square and staggered layouts show strong consistency across configurations ($\pm 1.5\%$ variation), the biomimetic arrangement systematically underperforms by 4%–6% under equivalent parameter sets. This discrepancy arises from the biomimetic layout’s inherent geometry: its mathematical design imposes a central open space, which locally increases inter-reactor spacing for a small subset of PBRs but forces tighter packing in peripheral regions. Consequently, while centrally positioned reactors achieve elevated view factors (≈ 0.21 – 0.22), the majority of reactors in densely packed outer zones experience significant shading (≈ 0.20). The numerical dominance of these peripheral units disproportionately lowers the system-wide average compared to the more uniform spatial distributions of square and staggered arrays. Although the biomimetic layouts exhibit a lower radiative efficiency, the modular scalability via

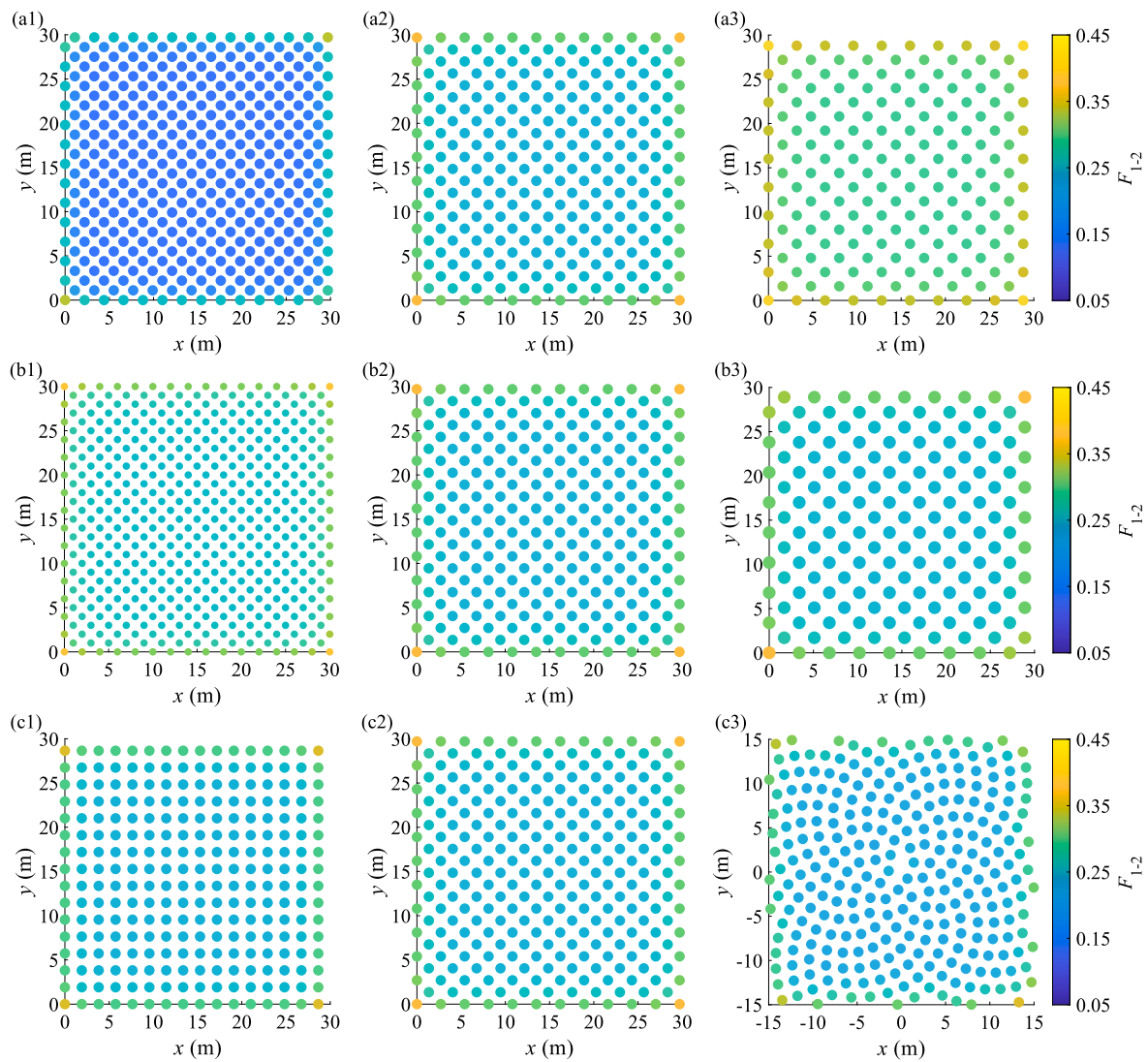


Fig. 6. View factors between cylindrical photobioreactor side surface and hemispherical sky, F_{1-2} , for different clearance: (a1) $s = 0.56$ m, (a2) $s = 0.91$ m, and (a3) $s = 1.26$ m in a staggered layout with baseline reactor radius $r = 0.5$ m; for different photobioreactor radius: (b1) $r = 0.25$ m, (b2) $r = 0.5$ m, and (b3) $r = 0.75$ m in the staggered layout with baseline spacing $s = 0.91$ m; for photobioreactors arranged in patterns of: (c1) square layout, (c2) staggered layout, and (c3) biomimetic sunflower-inspired layout, with baseline parameters of photobioreactor radius $r = 0.5$ m and spacing $s = 0.91$ m.

sunflower-like symmetry offers advantages for phased industrial expansion. Conversely, square/staggered configurations provide construction simplicity suitable for standardized mass production.

While this study establishes a view factor-based MCRT framework for diffuse radiation analysis, it should be noted that critical future extensions require an integration of direct solar dynamics and photoinhibition responsive growth kinetics. Subsequent research should combine radiative simulations with transient light-field analyses and biological productivity models, advancing towards holistic PBR optimization for industrial applications.

5. Summary and conclusion

In this study, the Monte Carlo ray tracing (MCRT) method has been employed to quantify radiative dynamics in outdoor large-scale photobioreactor (PBR) arrays, revealing radius and spacing as dominant parameters governing mutual shading. Central photobioreactors exhibited lower sky view factors than edge units due to omnidirectional occlusion. Conventional square (0.241) and staggered (0.242) layouts outperformed the sunflower-inspired design (0.226), where peripheral densification negated central void benefits. The inter-reactor spacings

exerted greater influence than the layout type, with sparse arrays ($s = 1.26$ m) achieving 84% higher average view factors than dense configuration ($s = 0.56$ m). The framework enables multiphysics optimization of industrial-scale algal cultivation systems.

E-supplementary data for this work can be found in e-version of this paper online.

CRediT authorship contribution statement

Yihan Zheng: Writing – review & editing, Writing – original draft, Resources, Investigation, Formal analysis, Data curation. **Lifeng Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Shengkai Dai:** Writing – review & editing, Formal analysis. **Donghang Liu:** Writing – review & editing, Formal analysis. **Hao Zheng:** Writing – review & editing, Data curation. **Song Yang:** Writing – review & editing, Formal analysis. **Yanming Guo:** Writing – review & editing, Formal analysis. **Muhammad Rafique:** Writing – review & editing, Formal analysis. **Bo Wang:** Writing – review & editing, Resources, Formal analysis. **Chi-Hwa Wang:** Writing – review & editing, Formal analysis. **Yong Shuai:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.biortech.2025.132813>.

Data availability

Data will be made available on request.

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